# AI and Data Science Professional

Course outline | 48 hours, 24 sessions of 2 hours | Updated 4 November 2026

## What it is

A live course that takes a working professional from an Excel spreadsheet to a tested model and a small web app. It follows one path in 24 sessions: think with data, Python for data, analyse and explain, machine learning, neural networks and text, then ship it and present. Every tool is free. Every dataset is fictional and set in Uganda (a retail chain, a microfinance lender, district clinics and customer messages, in shillings), so no real person or company is exposed. The tutor is Jotham Martin Wambi of KLAi Professional Learning.

## Who it is for

- **Working professionals in Uganda and East Africa** who handle tables and reports: analysts, accountants, credit and finance officers, programme and monitoring staff, ministry and NGO officers, managers.
- **Employers** who want staff to move from spreadsheets to data work they can check, explain and repeat, without buying software.

## Entry requirements

- **Confident in Excel or Google Sheets.** You can write a formula, sort and filter, and have seen a pivot table.
- **No code needed.** Python is taught from the first line, beside the spreadsheet idea it replaces.
- **No statistics needed.** Averages, spread, sampling and tests are taught with pictures and small examples.
- **A laptop and a free Google account.** Windows or Mac, with a browser and a charger. Nothing to install. A free GitHub account is needed from Session 23. No paid account, API key or graphics card is ever required.

## What you will be able to do

- Turn a business question into a decision, and check the data behind it before you report a number.
- Read a table by its records, variables and keys, then summarise and chart it fairly in a spreadsheet.
- Load, clean, group and join data in Python with pandas, keep a data cleaning log, and prove that a total reconciles to a control total.
- Ask a database the same questions in SQL, and check the answer against pandas.
- Explore a new dataset in a fixed order and write a finding with a number, a comparison and a limit.
- Test whether a difference between two groups is real, and fit and read a regression model.
- Build, test and judge classification, clustering and forecasting models against a baseline model, and catch data leakage.
- See how a neural network learns, turn text into numbers, and make a small assistant answer from your own documents, with a test and an audit log.
- Save a model, publish a small app, and check it for fairness and data protection before anyone relies on it.
- Present a result to a manager: the answer, the evidence beside a baseline, and the limits.

## How every session runs (2 hours)

| Starts | Minutes | What happens |
|---|---|---|
| **0:00** | 10 | Recall and the question of the day |
| **0:10** | 45 | Concepts: picture, plain words, one example by hand |
| **0:55** | 10 | Break |
| **1:05** | 45 | Lab: predict, check, write the dictionary |
| **1:50** | 10 | Exit check, keep and next |

Concepts come first. For 45 minutes the laptops stay closed while you see a picture, hear the idea in plain words, and follow one example by hand. Each idea is given its real industry name with a plain explanation beside

it (for example record, standard deviation, residual, decision threshold), so you can follow documentation, job interviews and later courses. Then you open the notebook or the sheet once and work for 45 minutes: predict the answer, run the cell, read the output, and change one thing. Automatic check cells tell you whether your numbers match. Every session ends with an exit check and a file or note you keep. Slides are shared as PDFs, written in full sentences, so you can revise without the tutor.

## The 24 sessions

### Module A: Think with data (Sessions 1 to 4)

| # | Session | Question of the day | Lab deliverable |
|---|---|---|---|
| 1 | What data science is and how to think in it | What does a data scientist do, and what will you be able to do after 24 sessions? | Notebook in Drive and your business question |
| 2 | Data as tables: records, variables and keys | What exactly is one row of a table, and how does one table connect to another? | Workbook dictionary tab: record, key, problems per tab |
| 3 | Summarising and showing data fairly | How do we describe thousands of records in one sentence and one chart without misleading anyone? | Workbook: three pivot tables and three charts |
| 4 | Uncertainty, samples and cause | How sure can we be of a number from a sample, and when does a pattern mean that one thing caused another? | Simulation tab: 200 samples of 30 orders |

### Module B: Python for data (Sessions 5 to 9)

| # | Session | Question of the day | Lab deliverable |
|---|---|---|---|
| 5 | Your first Python: variables, lists and error messages, next to the spreadsheet | What does the same price-list task look like in Excel and in Python, and how do you read an error message when Python stops? | Notebook: price list, one error read and fixed |
| 6 | Decisions, repeats and functions: if, for, and a tested function | How do we make Python decide and repeat, so we never copy a formula down thousands of rows? | Notebook: a tested loyalty_tier function |
| 7 | pandas I: load, look and filter a table of 30,229 orders | How do we open a file of 30,229 rows, look at it, and ask it the first questions? | orders_profile.csv: a data profile |
| 8 | pandas II: cleaning the mess, with a log and a reconciled total | How do we make a raw export trustworthy, and prove what we changed? | orders_clean.csv, cleaning log, reconciliation note<br>Project milestone 1 |
| 9 | pandas III: group, join and reshape | How do we answer a question that needs several groups and several tables, and trust the total? | Revenue tables and a join log with row counts |

### Module C: Analyse and explain (Sessions 10 to 13)

| # | Session | Question of the day | Lab deliverable |
|---|---|---|---|
| 10 | SQL for analysts: ask a database the questions you ask pandas | How do we ask a database the same questions in the language every employer uses? | Query book: ten SQL queries with pandas proofs |
| 11 | Exploring and visualising data with Python: from a new file to a finding | How do I explore an unfamiliar dataset in one sitting and find what is worth reporting? | Notebook: eight charts and three findings |
| 12 | Inference: is the difference real, or could it be luck? | When two groups differ, how do I know the gap is real and that chance does not explain it? | Notebook: the offer test five ways, a decision note |
| 13 | Regression: explaining and predicting a number | How much does one thing change when another changes, and how well can a model predict it? | regression_summary.csv and a one-page report<br>Project milestone 2 |

## Module D: Machine learning (Sessions 14 to 19)

| # | Session | Question of the day | Lab deliverable |
|---|---|---|---|
| 14 | Machine learning: the workflow, from loans data to a first model | What does it mean for a computer to learn from examples, and how do we know it did? | loans_prepared.csv and a three-sentence note |
| 15 | Classification: predicting a yes or no answer with logistic regression and a decision tree | How do we predict a category, and what does the model's confidence mean? | loan_risk_scores.csv and a threshold note in UGX |
| 16 | Judging a model: metrics, leakage and imbalance | How do we tell a model that works from one that only looks good? | model_comparison.csv and a note on the leaked column |
| 17 | Forests, boosting and overfitting: a stronger model without fooling ourselves | How do we get a stronger model without fooling ourselves? | Model comparison table and a decision memo |
| 18 | Finding structure without labels: customer segments and unusual orders | How do we group customers, or spot odd cases, when nobody gave us the answers? | customer_segments.csv and unusual orders to review |
| 19 | Forecasting and a first end-to-end model: baseline, time-based split and Pipeline | How do we predict next month, and how do we pack a whole modelling job into one object we can trust? | Baseline, tested forecast, Pipeline, manager note<br>Project milestone 3 |

## Module E: Neural networks and text (Sessions 20 to 22)

| # | Session | Question of the day | Lab deliverable |
|---|---|---|---|
| 20 | Neural networks and deep learning, without the maths wall | What is a neural network, and when is one the right tool? | Notebook, results table and a recommendation |
| 21 | Text as data and meaning as numbers | How can a computer compare customer messages when it only understands numbers? | feedback_scored.csv and a three-sentence finding |
| 22 | Language models and building with them | What does a language model do, how do we test one, and how do we make it answer from our own documents? | eval_results.csv and audit_log.csv |

## Module F: Ship it and present (Sessions 23 to 24)

| # | Session | Question of the day | Lab deliverable |
|---|---|---|---|
| 23 | Ship it: reproducible work, a small app and responsible use | How do we turn a notebook into something a colleague can repeat, use and trust? | Shipping folder: app.py, saved model, model card |
| 24 | Present your course project and plan your path from here | Can you explain what you built, what it can and cannot do, and what you will learn next? | Presentation sheet and a six month learning path<br>Course project talk |

Sessions 1 to 4 and the planning half of Session 24 are spreadsheet work with no code. Sessions 5 to 23 have a Python notebook for Google Colab.

## Tools: all free

| | |
|---|---|
| **Spreadsheets** | Google Sheets, or Excel, or LibreOffice Calc |
| **Python in the browser** | Google Colab, with pandas, NumPy, SciPy, statsmodels, scikit-learn, matplotlib and seaborn |
| **Databases** | SQLite, run inside Python |
| **Neural networks and text** | TensorFlow Playground in the browser; sentence-transformers (all-MiniLM-L6-v2) run inside Colab |
| **Sharing and shipping** | Git and GitHub; Streamlit Community Cloud for a small public app |
| **Optional help** | Any free chat assistant you already use. Every lab runs without one. |

Free services change their limits. Where the course quotes a limit it says so, and tells you to check the current terms.

## What your employer gets

Each session ends with a task you can do at work the next Monday. A few examples taken from the sessions:

- Clean any raw export, log each change, and reconcile the total to a control total.
- Query a company database with SQL and prove the answer in pandas.
- Test whether a gap between two groups is real, report its size and interval, and avoid false alarms.
- Score each loan, customer or transaction with a probability, then set the decision threshold from what each mistake costs.
- Make an assistant answer from your own documents, test the search, and keep an audit log of AI-assisted work.
- Make a model repeatable, shareable and checkable for fairness and data protection before anyone relies on it.

## No exams, no marks: practical projects you keep

- There is no exam and no assessment of any kind. No marks, grades, pass marks or weights. You build practical projects and you keep them.
- Each lab has automatic check cells that print OK or NOT YET, so you see at once whether your numbers match. The check cells, the recall questions and the exit-check questions are self-checks. Nobody marks them.
- Three project milestones: Session 8 (a cleaned file with its log and a reconciliation note), Session 13 (a one-page analysis report) and Session 19 (a tested model beside a baseline model). You show each to your tutor for feedback.
- The course project: you scope a question in Session 13, choose a dataset in Session 14, test a model in Session 19, publish a small app and model card in Session 23, and present for ten minutes in Session 24. The Course Projects Guide has the details and a project checklist for feedback.

## Certificate and what you keep

- A certificate of completion from KLAi Professional Learning, given for completing the course project. There are no criteria beyond that.
- A project from your own question: your course project notebook, a small app and model card on your own GitHub, and your ten-minute talk. Most learners use a course dataset. If your employer agrees in writing and the data holds no personal details, you may use work data.
- A folder of files from the 24 labs (cleaned data, query book, charts, model comparison, scored files) that you can show an employer.
- An external credential or university accreditation is not part of the course.

## What the course does not cover

Forty-eight hours is short, so the course teaches the working core and says where it stops. These are covered in Session 24 with a free next step for each.

- Mathematics by hand: You calculated results by hand but did not prove them.
- Data structures and algorithms: Interview puzzles need separate practice.
- Big data tools: Your data fitted in one notebook.
- Paid cloud platforms: AWS, Azure, Google Cloud. You used free tiers of Colab, GitHub and Streamlit.
- Deep learning from scratch: One neuron and a small network in numpy. The pretrained model was optional.
- Dashboards and BI tools: Power BI and Tableau were named, not taught.
- Production MLOps: No container, automated tests or monitoring.

Dates and fees are on the course flyer.